產品開發策略

產品設計/**製造**/銷售

產品開發策略簡介

產品管理的實用手法

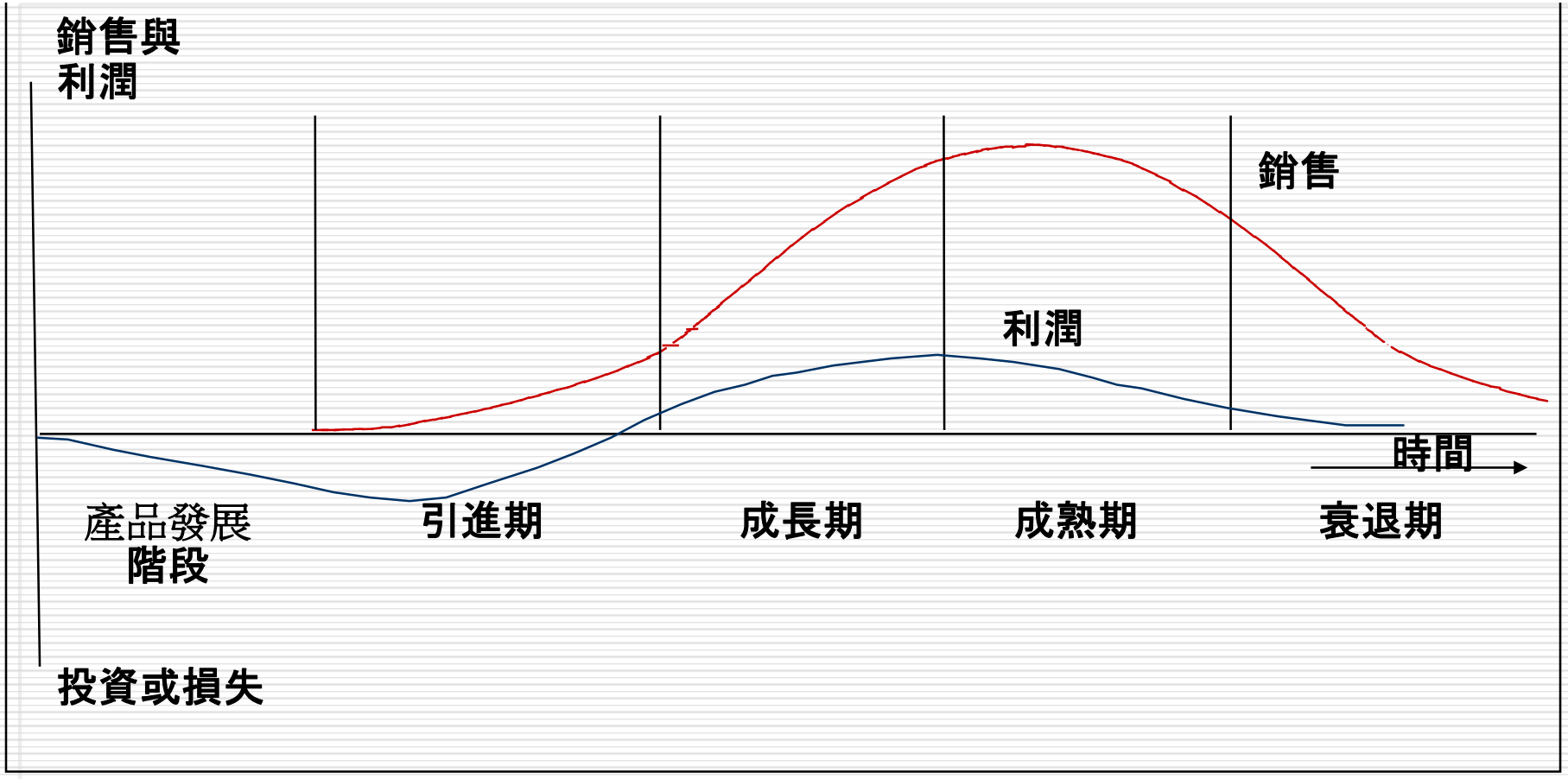
QFD/Kano Model

產品開發策略

- 規劃
  - 目標市場、企業目標、主要的假設與限制條件
- 概念設計
  - 定義市場需求、發展評選產品概念(外觀、特色)方案(評估投資、市佔率、產品壽命週期、產品線幅度)
- 系統整體設計
  - 定義整體架構與組成元件、生產系統之組成要素
- 細部設計
  - 定義零件型態(工程圖)、物料表、品質規格、製程及對應之設備及工具
- 測試與修正
  - 發展產品雛型以測試品質規格達成狀況
- 試產
  - 以擬採行之生產系統製作新產品，訓練人員、發現與解決潛在問題

產品生命週期管理(PLM)

Sales and Profits Over the Product Life



產業生命週期分析

1.早期市場：應用研究、技術創新

2.利基市場：產品創新、維持毛利

3.全面市場：主導標準、擴大市佔

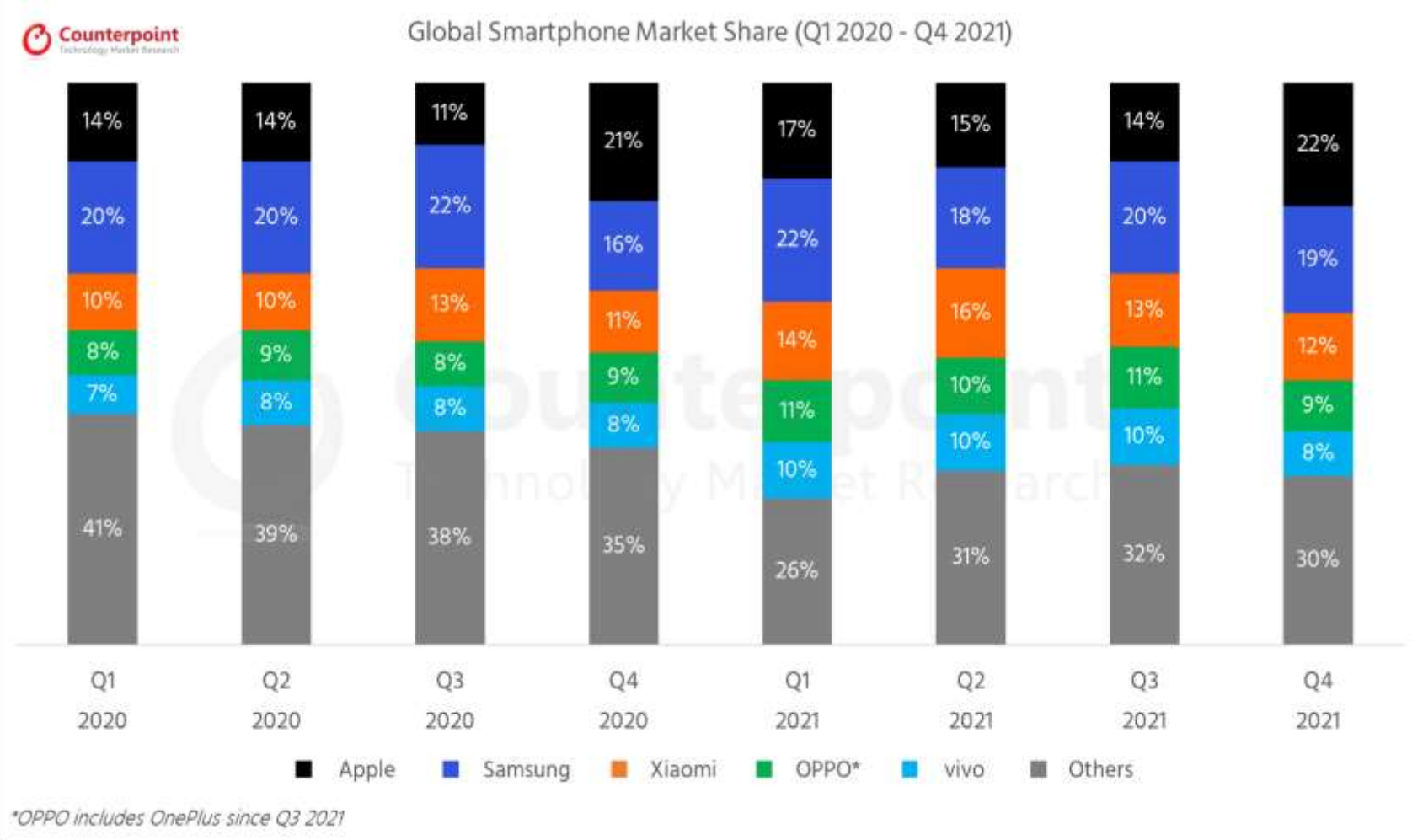
4.成熟市場：製程創新、Cost down

5.衰退市場 ：尋找另一個應用出現

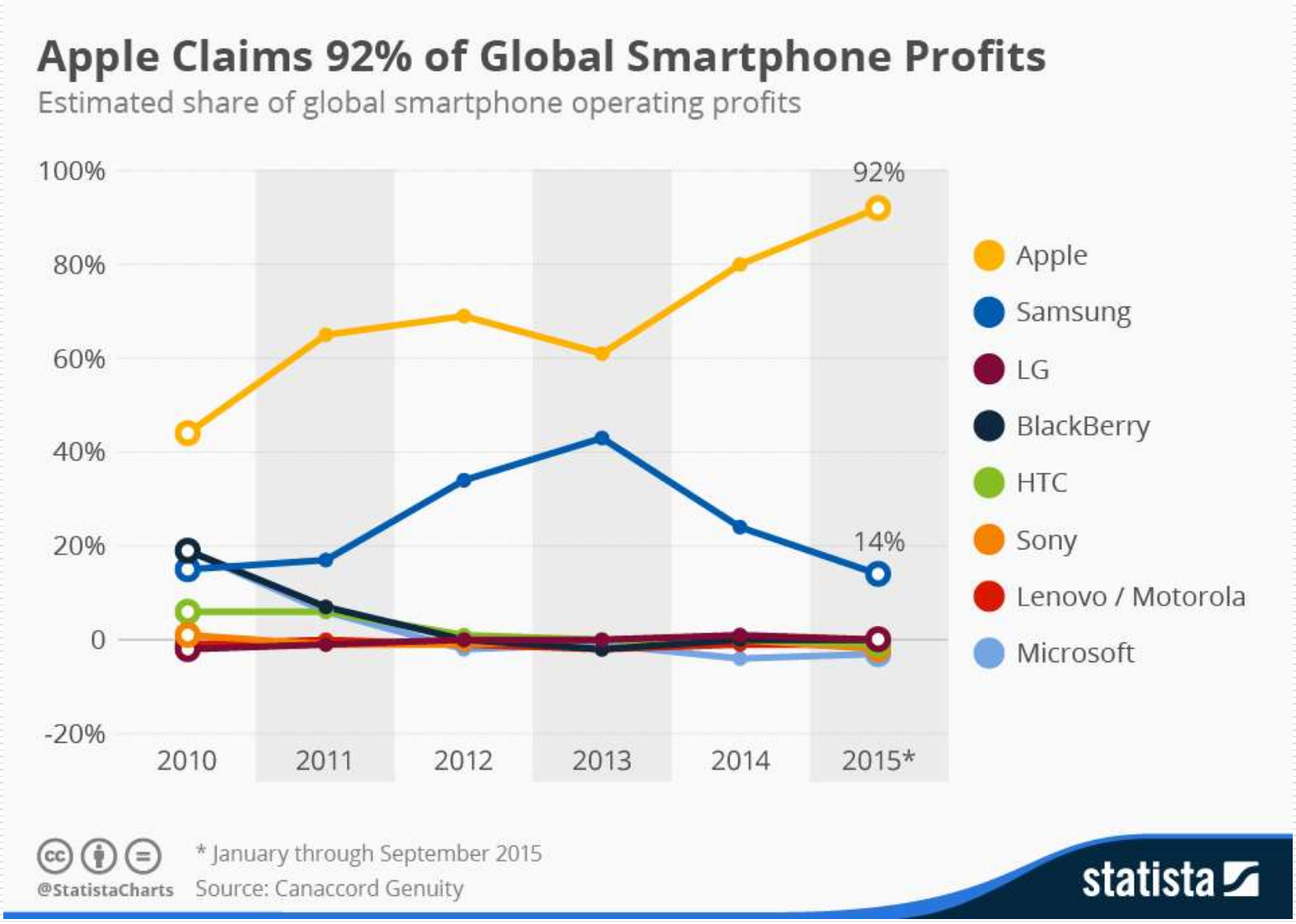
新產品失敗之原因

- 產品設計缺陷
- 上市時間不對
- 競爭者太強
- 市場分析錯誤
- 成本計算錯誤
- 產品定價錯誤
- 產品定位錯誤
- 配銷通路不當
- 廣告促銷不當

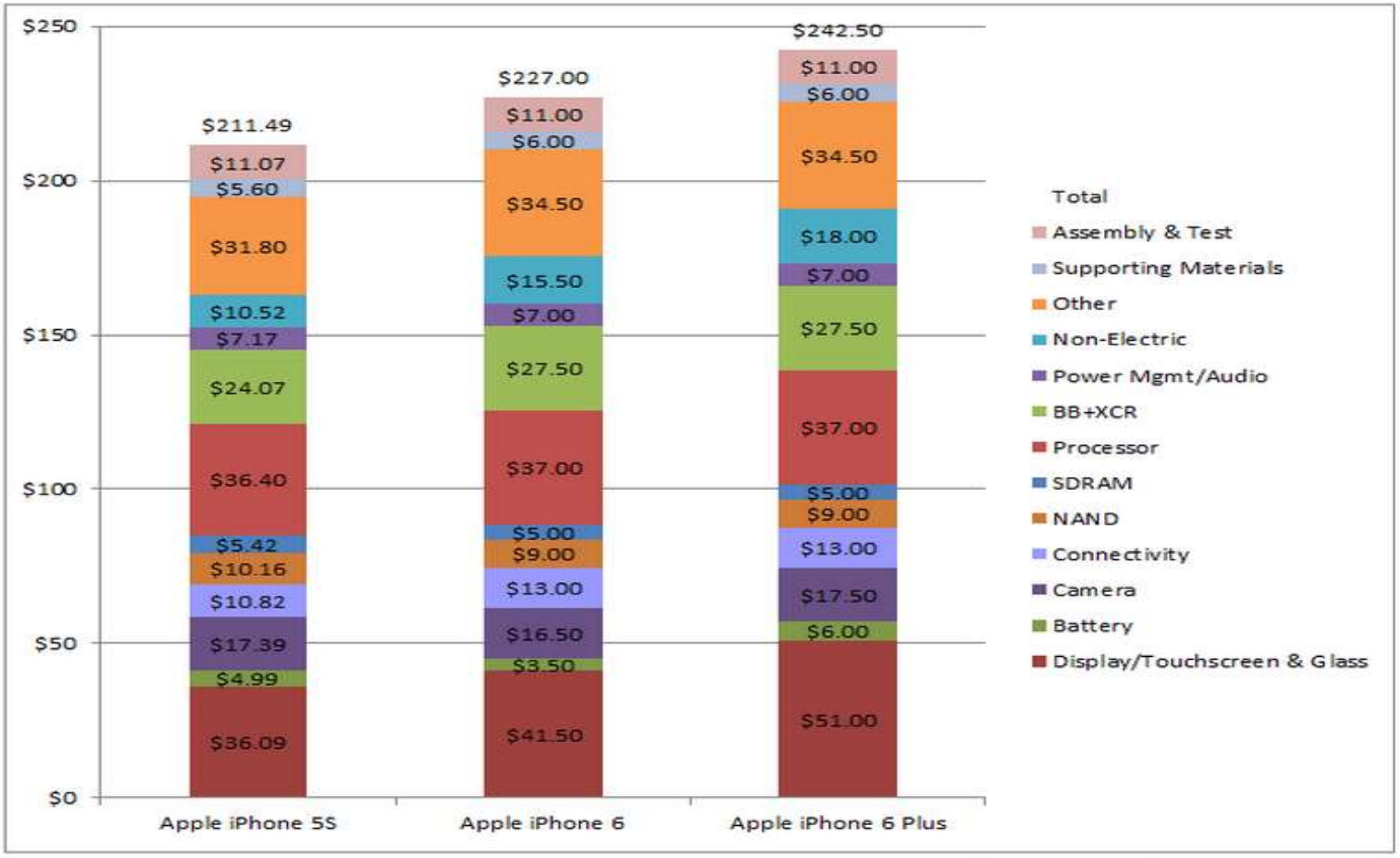
Smart phone's market share



Smart phone's profit share



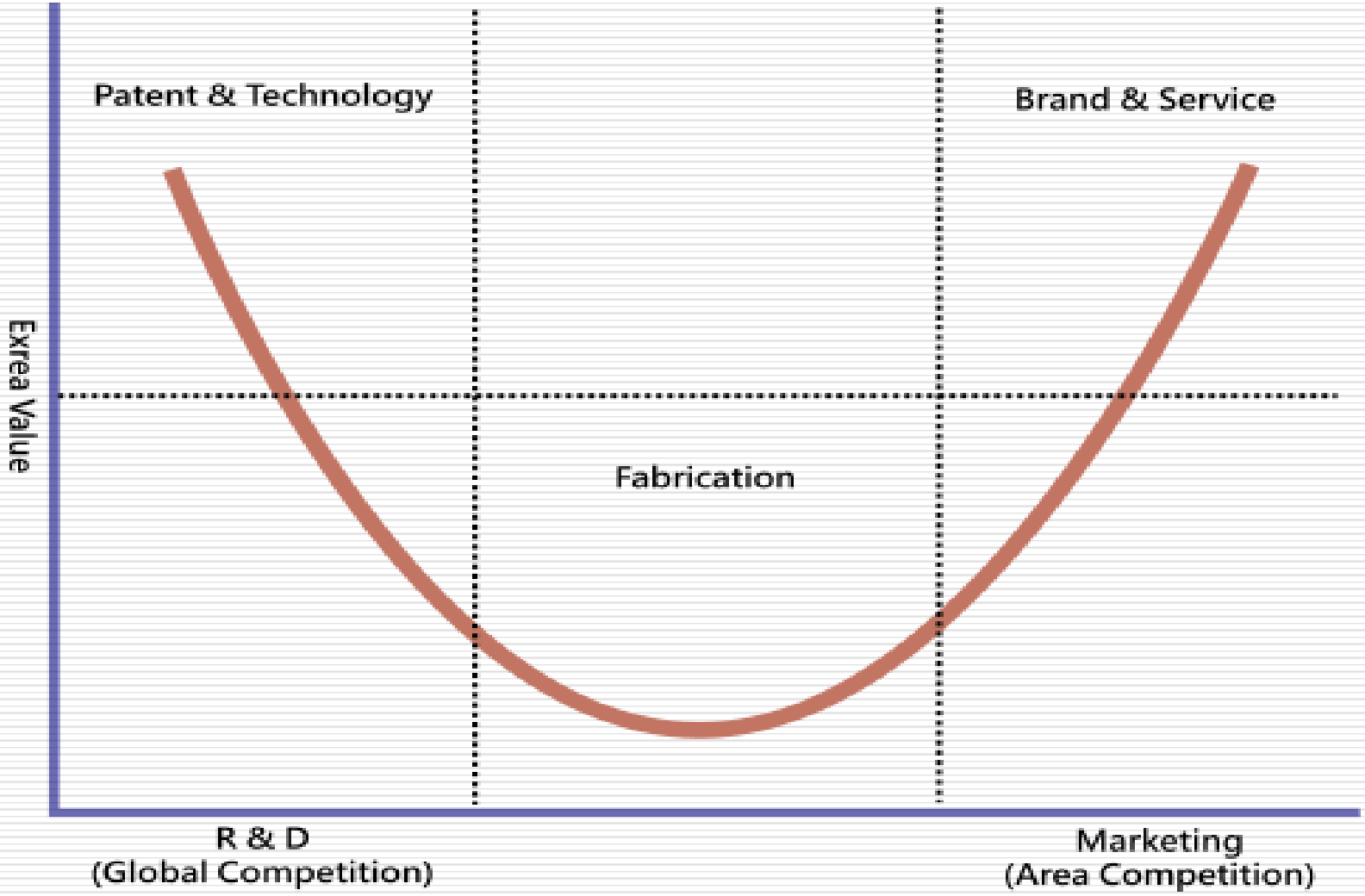
Apple iPhone's cost structure



teardown.com

Source: http://www.forbes.com/sites/chuckjones/2014/09/24/apples-iphone-6-teardown-and-other-costs-analysis/#41f124b72327

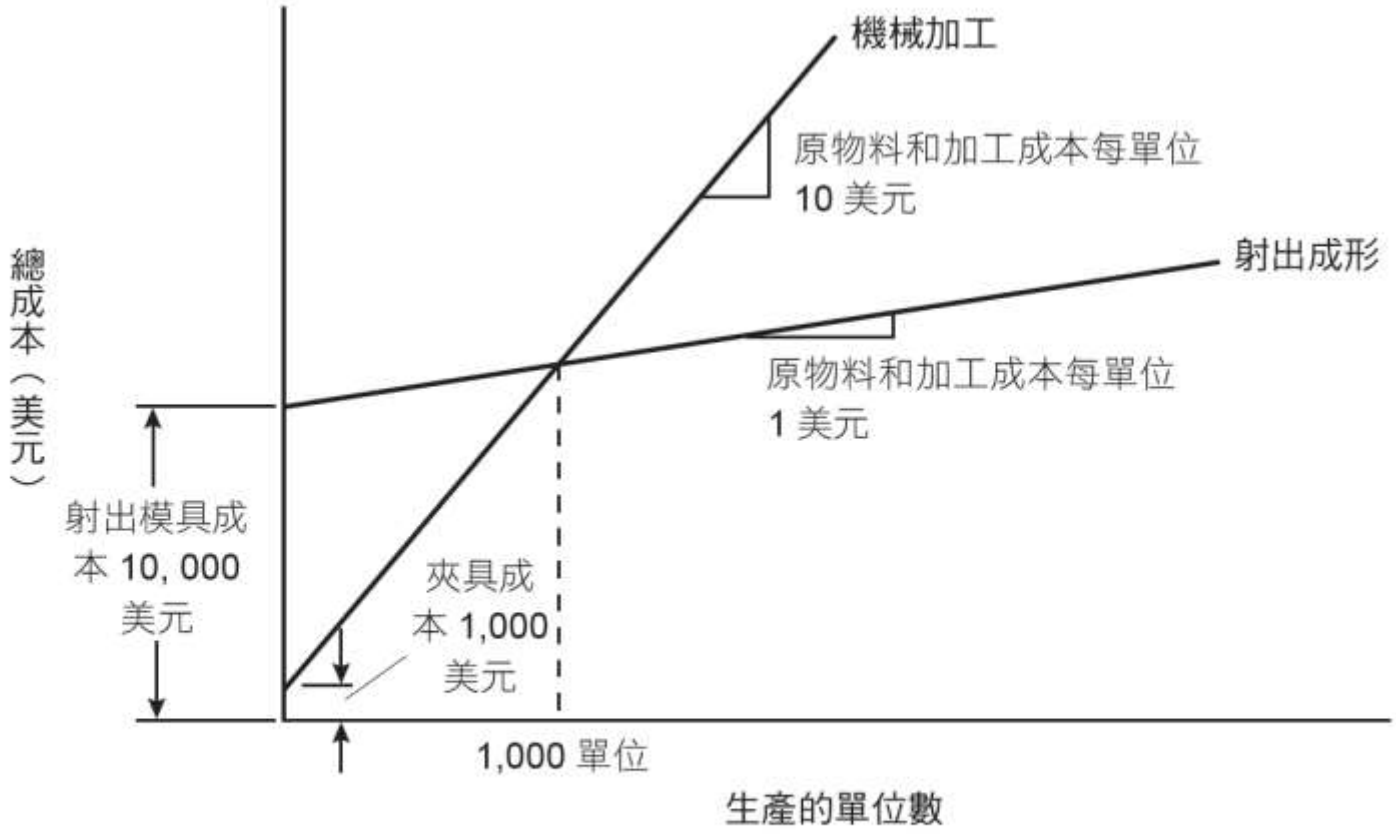
Acer- Sze's Smiling Curve (微笑曲線)



Video 1

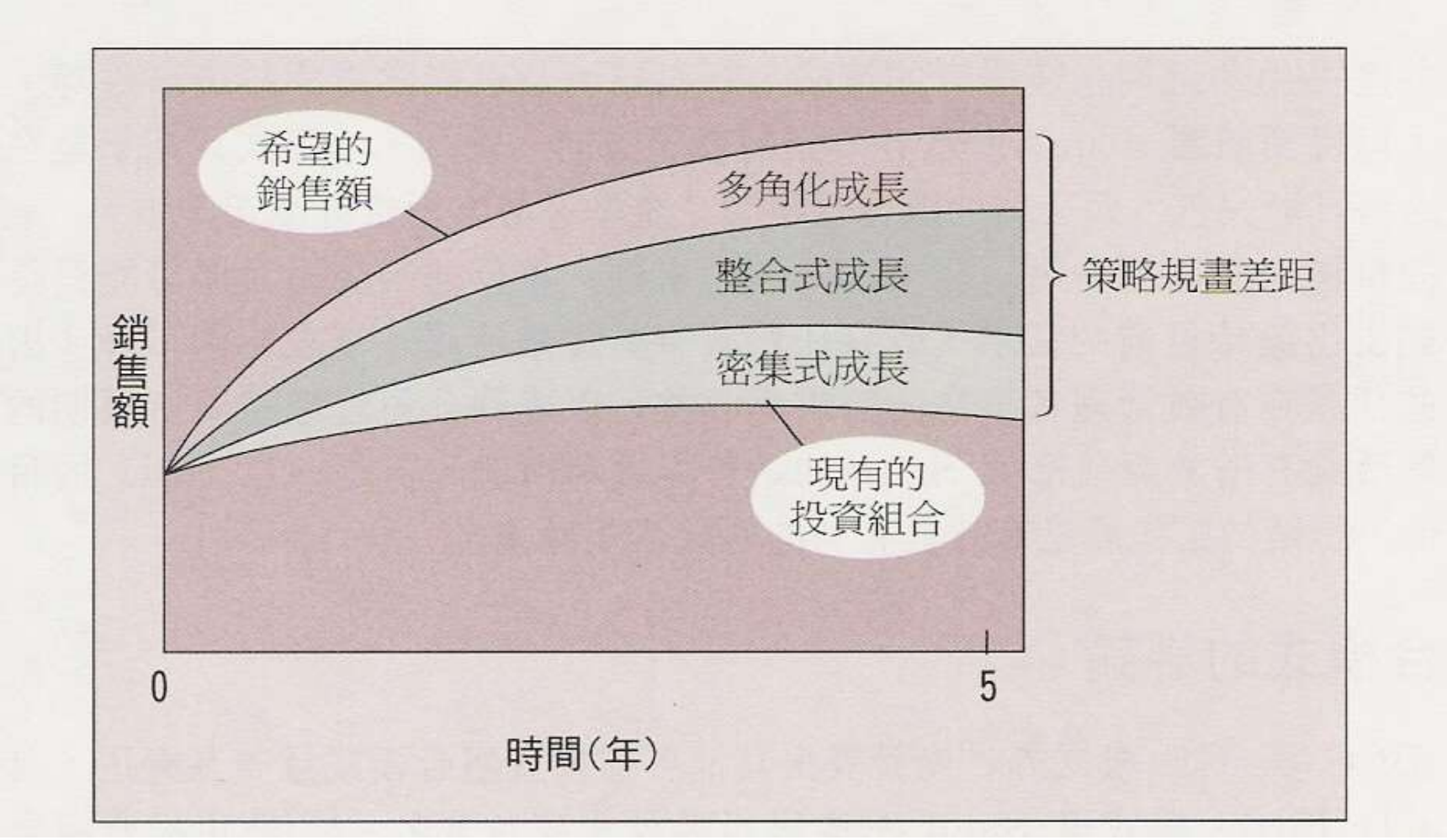
- *A short video to tell you Taiwan's innovation capabilities in 2004*
- *What new product/service is promising now?*
- *What resources do you have?*
- *Whom do you want to sell?*
- *How to promote your product/service?*
- *How long does it take to break even?*

Break-even analysis



圖表 13-10 射出成形與機械加工的單位成本假設之比較。

策略規劃缺口



填補策略規畫差距

密集式成長機會

在公司現有的事業中尋求進一步成長的機會。

整合式成長機會

建立或取得與公司現有事業有關的事業機會。

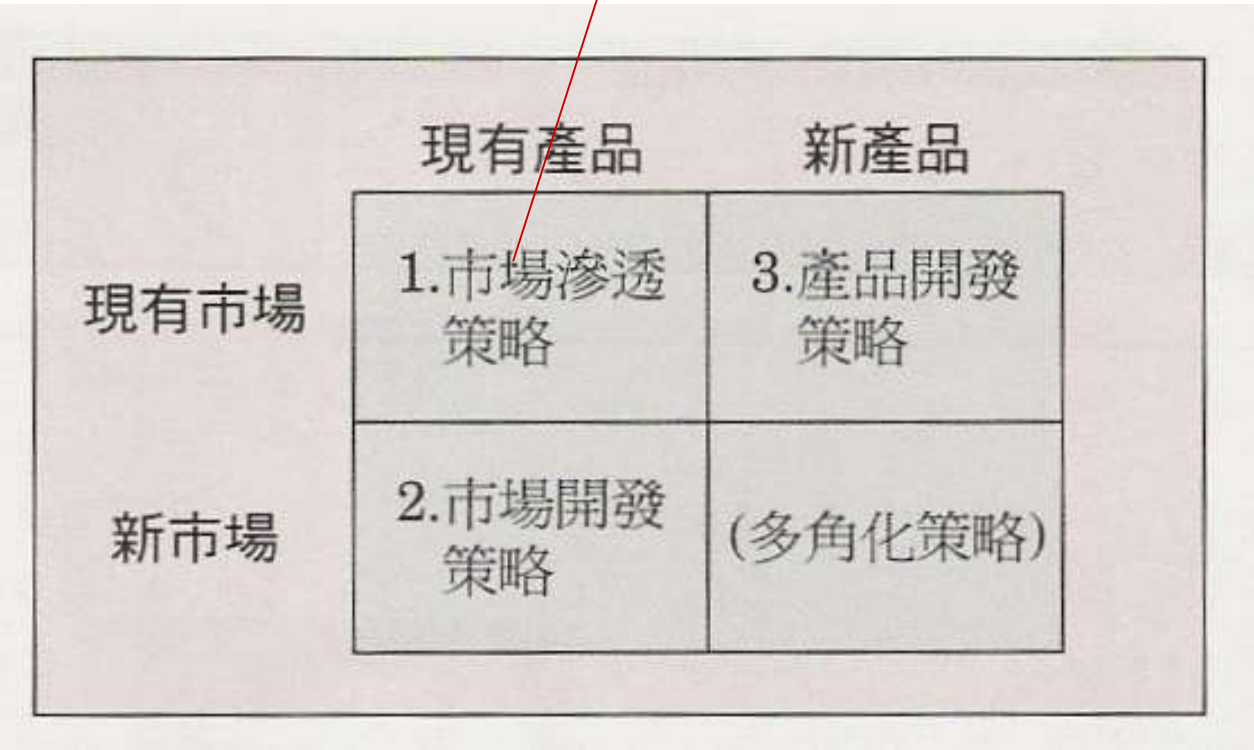
多角化成長機會

增加具吸引力且與現有事業不相關的事業機會。

填補策略規畫差距一

每朝健康綠茶提出「三比八」黃金身材比例的訴求，鼓勵消費者天天喝每朝。

密集式成長機會



填補策略規畫差距一

密集式成長機會



每朝健康進軍大陸市場

	現有產品	新產品
現有市場	1.市場滲透策略	3.產品開發策略
新市場	2.市場開發策略	(多角化策略)

填補策略規畫差距一

密集式成長機會



開發新產品「每朝健康烏龍茶」

	現有產品	新產品
現有市場	1.市場滲透策略	3.產品開發策略
新市場	2.市場開發策略	(多角化策略)

填補策略規畫差距二

整合式成長機會

- 建立或取得與公司現有事業有關的事業機會
  - 向後整合：康師傅併購全家便利超商。
  - 向前整合：康師傅併購乾燥蔬菜製造廠。
  - 水平整合：康師傅併購味丹泡麵與味全品牌。

填補策略規畫差距三

多角化成長機會

- 增加具吸引力且與現有事業不相關的事業機會：
  - **集中式多角化**－不同顧客但產品線具有綜效，如華碩製造主機板與電腦(PC/NB)。
  - **水平式多角化**－現有顧客但產品線不相關，如永豐餘投資電子紙市場(元太科技)。
  - **複合式多角化**－與現有產品或市場皆無關，如台達電或台積電進入綠能領域。

成長機會分類整理

密集式成長	整合式成長	多角化成長
市場滲透	上游(向前)	集中式
市場開發	下游(向後)	水平式
產品開發	水平	複合式

全聯買下大潤發，公平會有條件通過！超市、量販將採雙品牌營運｜數位時代 BusinessNext (bnext.com.tw)

鴻海斥資25.2億元買下旺宏6吋廠、聚焦車用半導體！對雙方各有哪些好處？｜數位時代 BusinessNext (bnext.com.tw)

鴻海大手筆入股紫光，背後深藏兩大盤算 - 今周刊 (businesstoday.com.tw)

What are new products?

1. Promoting/exporting current products to new markets
2. Radical innovation like using magnetic RAM to replace dynamic RAM,
3. Process improvement to increase yield rate or lower costs (OLED displays, solar-cells' efficiencies)
4. Extension of existing products- from digital cameras to smart cameras (i.e. Wifi),
5. Repositioning old products- changing medical drugs (for therapy) into healthcare drugs (for prevention)
6. Product-service system (PSS): car dealers switch to rental & maintenance service by using IoT sensors

TRIZ (the theory of inventive problem solving)

Inventor : Genrich Altshuller (Russian)

Location : Russian researchers

Analysis 200,000 patents to find a rule

Text mining + Patent analysis??

Anti-wind umbrella– Spear VS. Shield
(#4 Asymmetry)



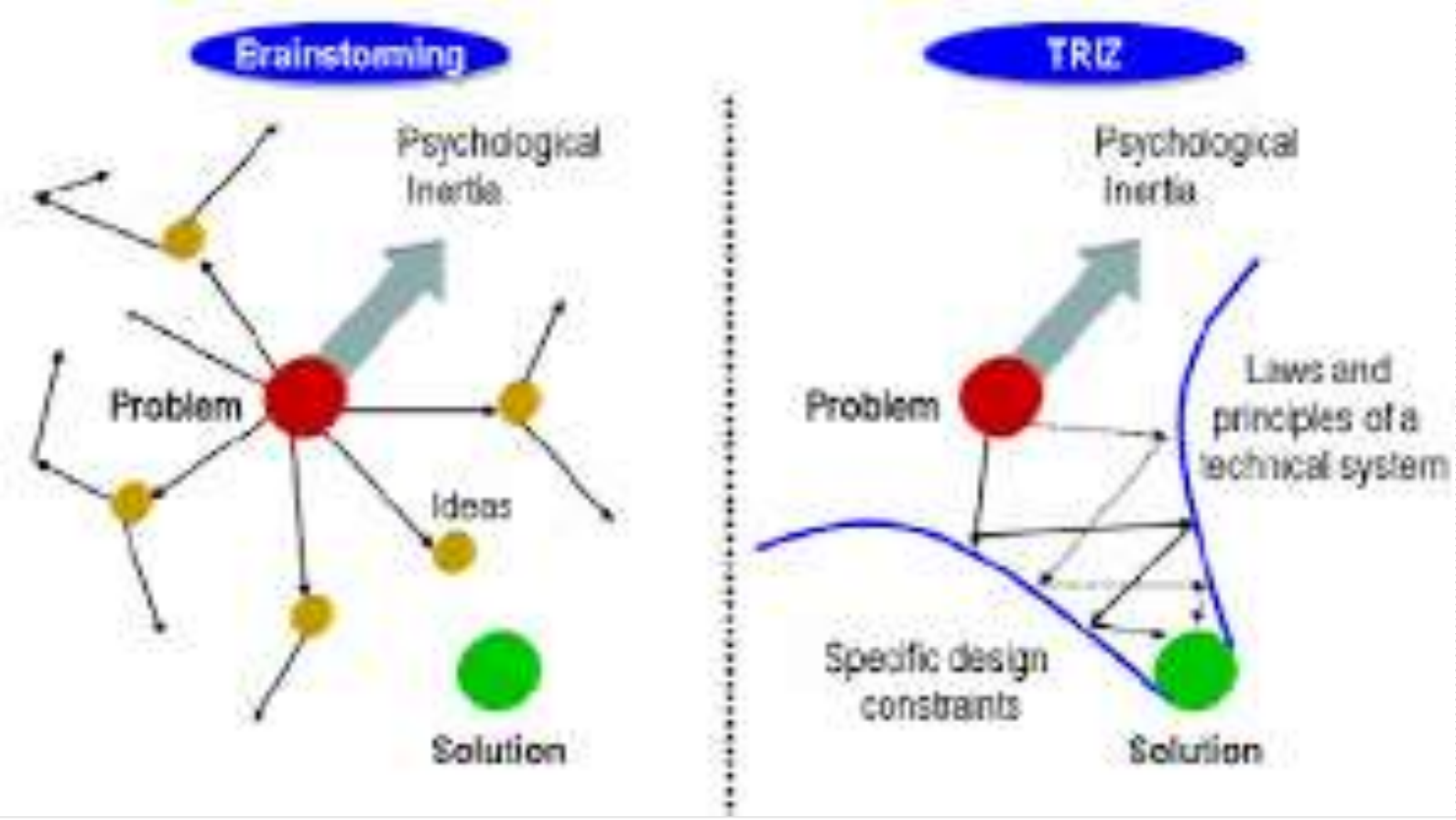
Genrikh Altshuller

October 15, 1926 - September 24, 1998

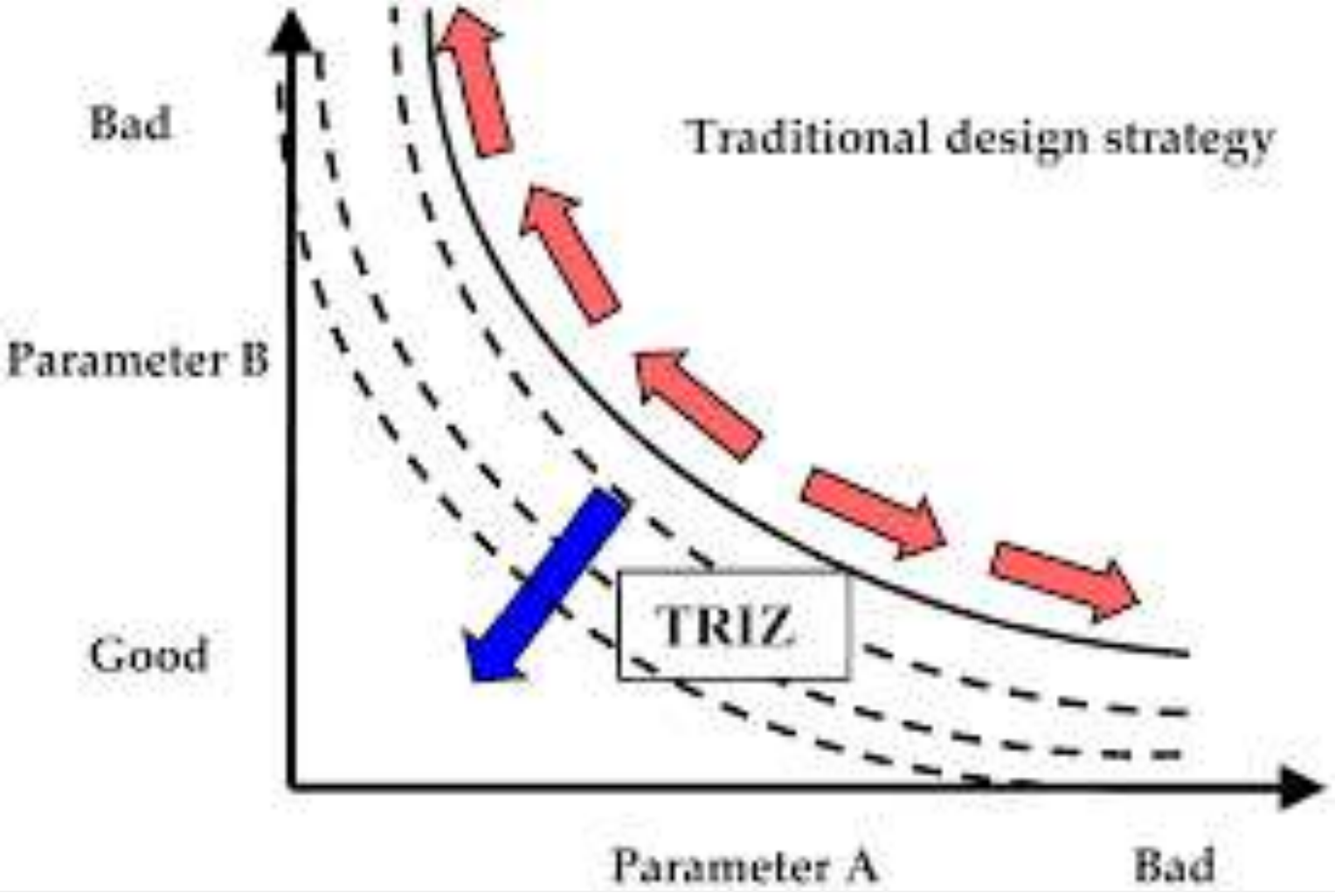
(467) 智慧機器人輪椅 貼心照護銀髮族 - YouTube

(467) SENZ Umbrellas - Highlights - YouTube

TRIZ's general process



TRIZ's inventive solution



TRIZ's design for phone-cameras



#1- Sony's QX series



#5- Samsung's *K* zoom



#28- Nokia's Lumia 1020

TRIZ's design for phone-pads



#7- Asus's PadFone



#14- LG's G flex



#17- Samsung's galaxy beam

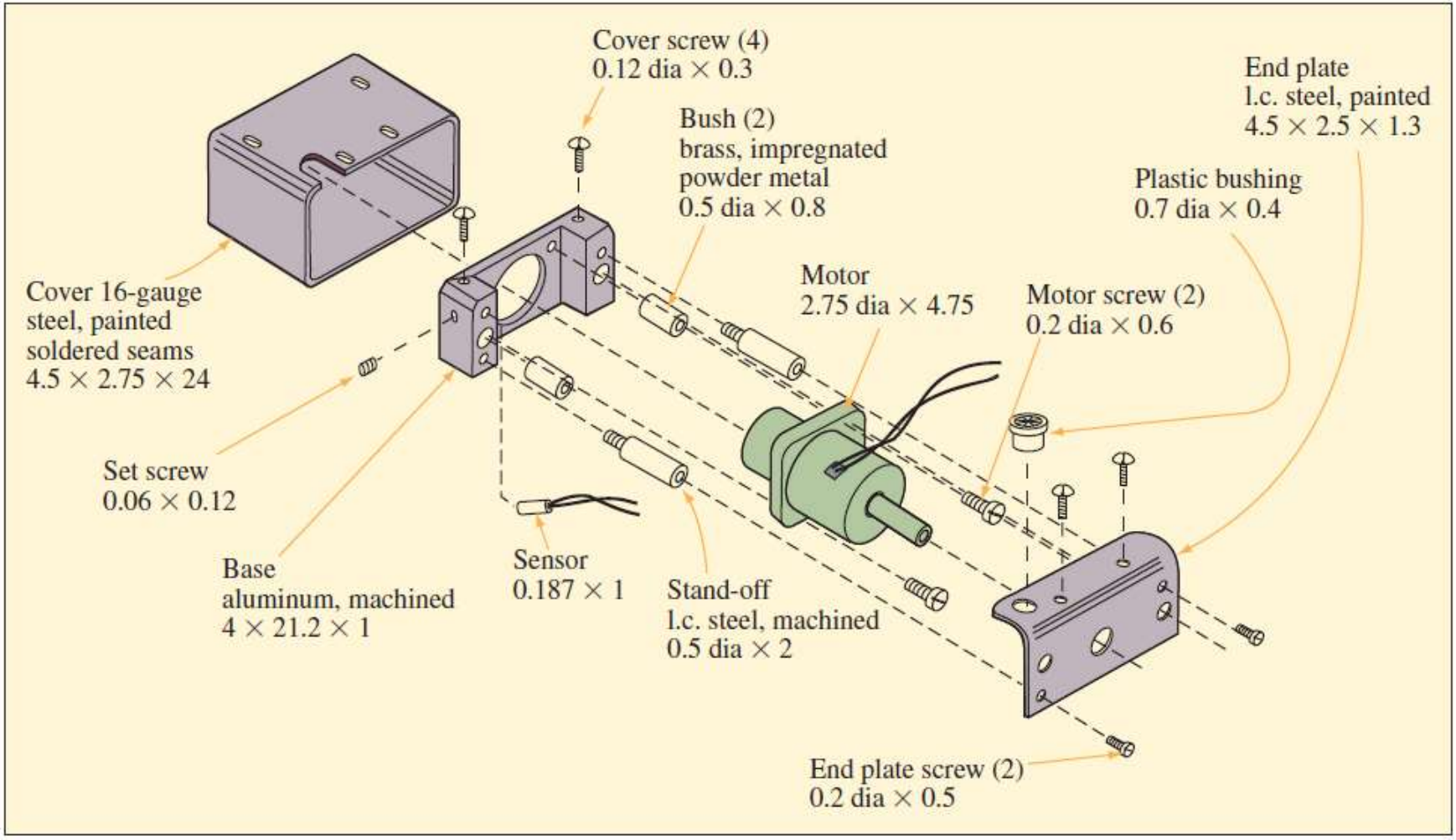
Design for Manufacturing (DFM) and Assembly (DFA)

- 符合生產的要求
  - 為製造而設計
    - 產品設計時考量到與組織製造能力(產能、裝配能力等)的相容性
    - 易製性
      - 簡單容易的設計原則
      - 設計階段即考慮製造問題
  - 為裝配而設計
    - 產品設計時考量到產品如何裝配
    - 減少裝配線上零組件的數目、裝配方法及裝配的程序

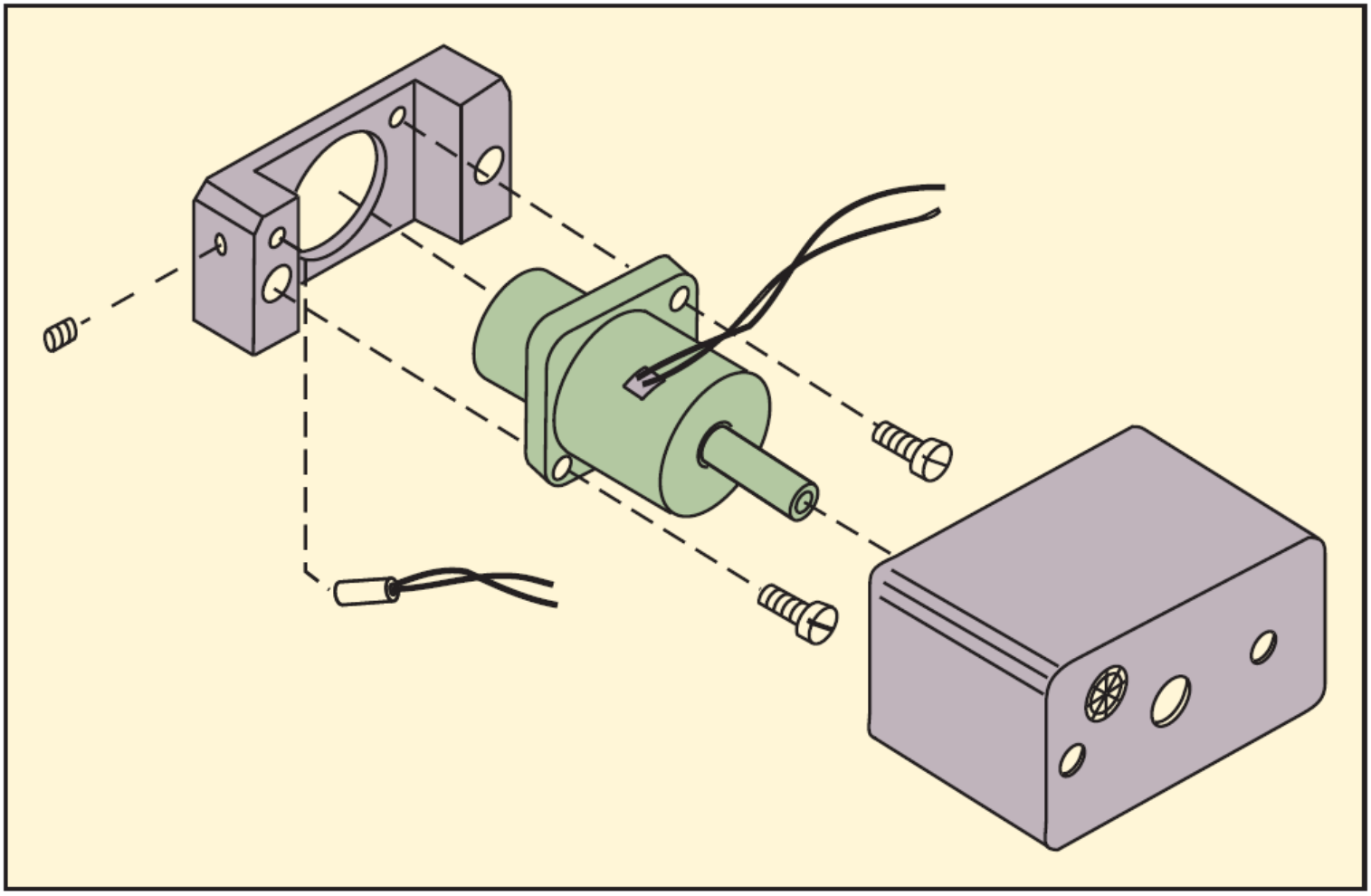
Industrial Practices

- 為製造而設計或為裝配而設計其目的都為了降低生產成本
  - 儘量以最少的零件來設計產品
  - 最底層零件能盡量作為其他零件組裝時的基座
  - 以按扣方式取代螺絲、螺帽等裝配方式
  - 零件之順序必須考量，以避免裝上後必須取下重裝
  - 零件應容易握持與定位
  - 利用工作研究方法決定作業員動作的經濟原則
  - 使用標準化且易得的更換零件
  - 避免過度客製化要求超出需要的公差

Proposed Motor Drive Design



Redesign of Motor Drive Assembly Following DFM/DFA



Ecodesign

- **Ecodesign**: the incorporation of environmental considerations in the design and development of products or services
  - ✓ The whole life cycle is considered
  - ✓ The product is considered as a system
  - ✓ A multi-criteria approach is used
- Application of ecodesign can benefit business



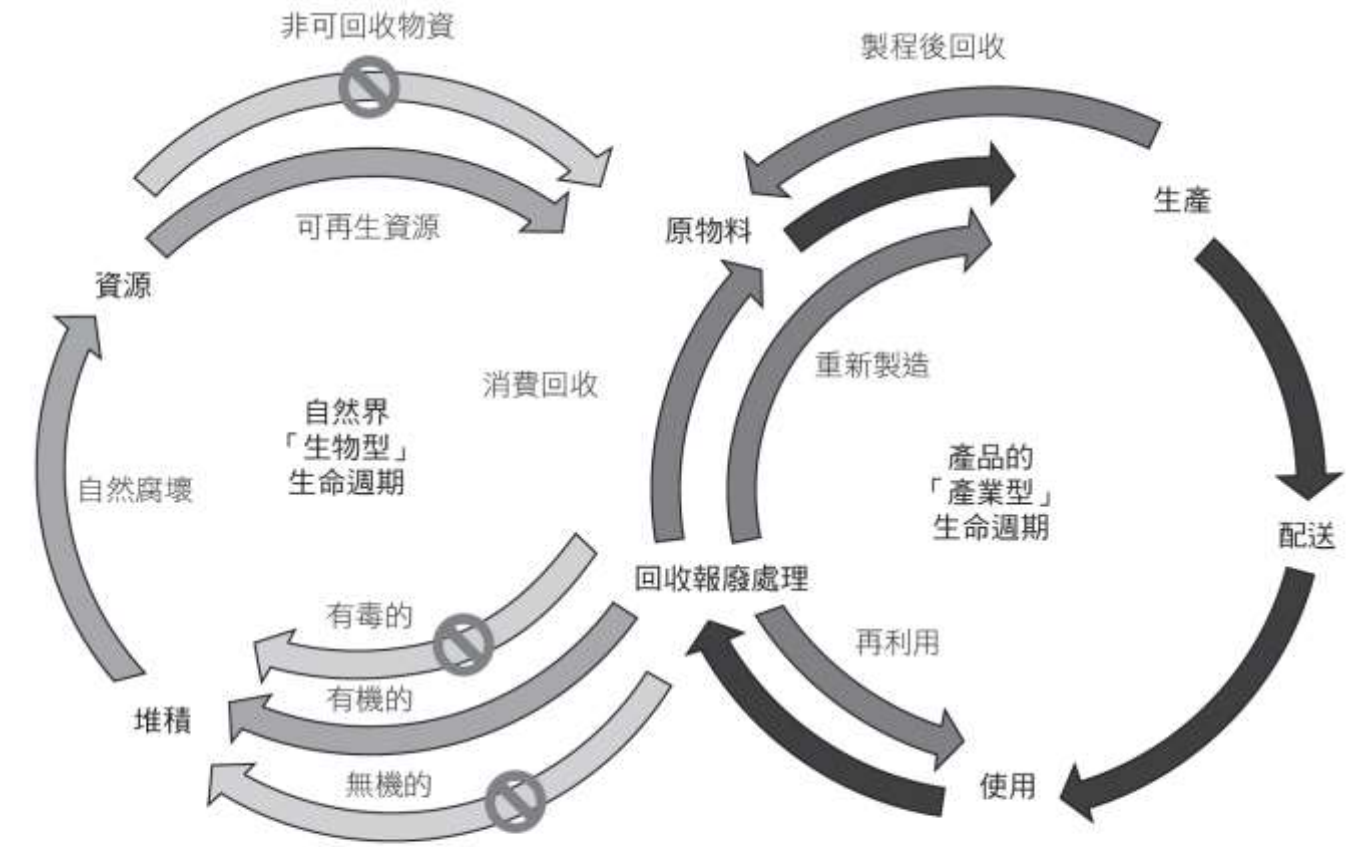
BioBag biodegradable and compostable plastic bags are used for carrying produce and are supplied on a roll. Text on the bag says “This bag is certified compostable. Use, reuse, then compost.”

Environmental impacts of product development

環境衝擊

- 全球暖化 (global warming)
- 資源逐漸耗盡 (depletion of natural resources)
- 固態廢棄物 (solid waste)
- 水污染 (water pollution)
- 空氣污染 (air pollution)
- 土地退化 (land degradation)
- 生物多樣性 (biological diversity)
- 臭氧逐漸耗盡 (depletion of ozone layers)

Nature lifecycle and product lifecycle

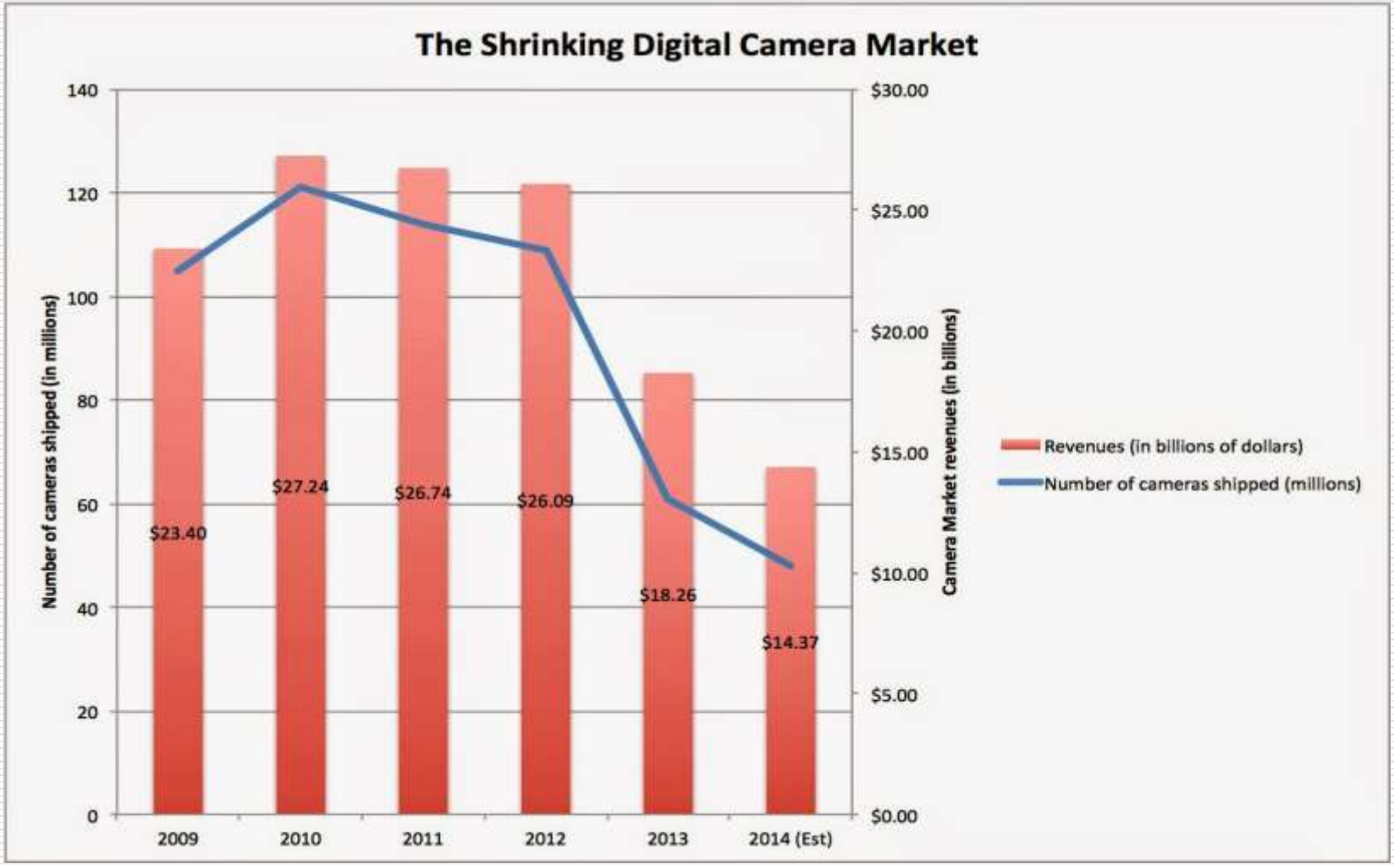


圖表 12-3 自然生命週期和產品生命週期。

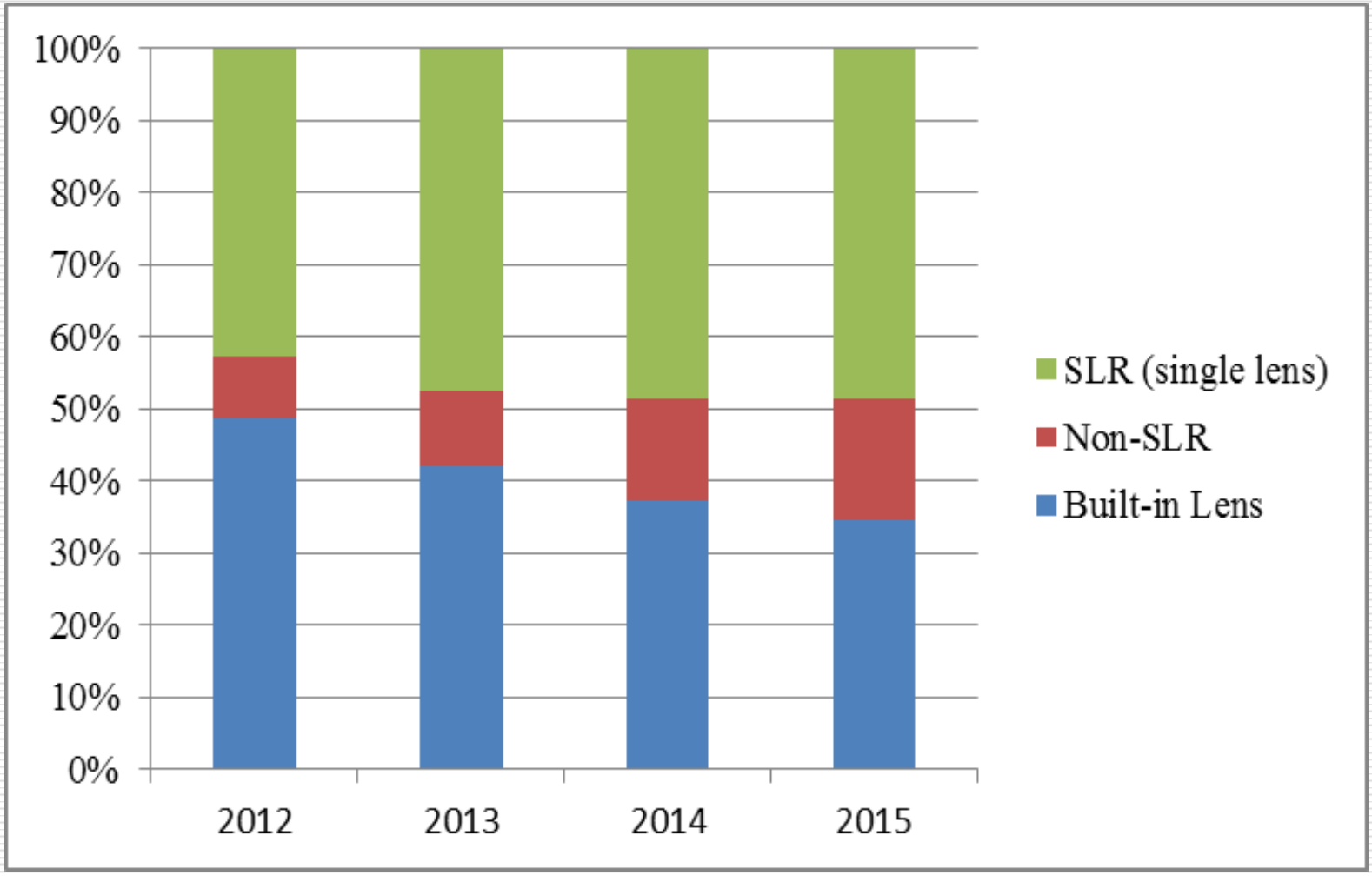
新產品開發的步驟

- Opportunity identification
- Concept generation & evaluation
- Prototype development
- Prototype testing & refinement
- Production management (resource allocation)
- Commercialization & launch management (after-sales)

A declining trend of digital still cameras



Worldwide camera's shipment value



Camera's product characteristics

	Specifications (levels)	Descriptions
Affective features (multi-levels)	F1 Zoom lens (3)	Standard (F11), Exchangeable (F12), Periscopic (F13)
	F3 Screen type (3)	Dual (A21), Up-twistable (A22), Side- twistable (A23)
	F3 Flash light (3)	Embedded (F31), Pop-up (F32), Add-on (F33)
Functional attributes (dichotomous)	A1 Big image sensor	Greater than 2/3 inch
	A2 Super wide angle	Less than 25mm
	A3 Ultra optical zoom	More than 10X
	A4 High-speed shutter	Above 10fps
	A5 Large aperture	Below f=3.0
	A6 Water proof	At least 5m
	A7 Dust proof	At least IP67
	A8 Digital compass	Direction identification
	A9 GPS tagging	Position tracking
	A10 Wifi connectivity	Internet communication
	A11 Touch panel	High sensitivity
	A12 Electronic view finder	Better visualization under exposure of sunlight

Kansei feature 1- Zoom Lenses



Exchangeable lenses-
Sony's A6000



Built-in lenses-
Sony's WX300



Periscopic lenses-
Sony's TX30

Kansei feature 2- screen type



Dual screen-
Samsung's DV300f

Up-twistable screen-
Samsung's MV900f

Side-twistable screen-
Samsung's EX2

Kansei feature 3- flash light



Embedded flash-
Nikon's S6900

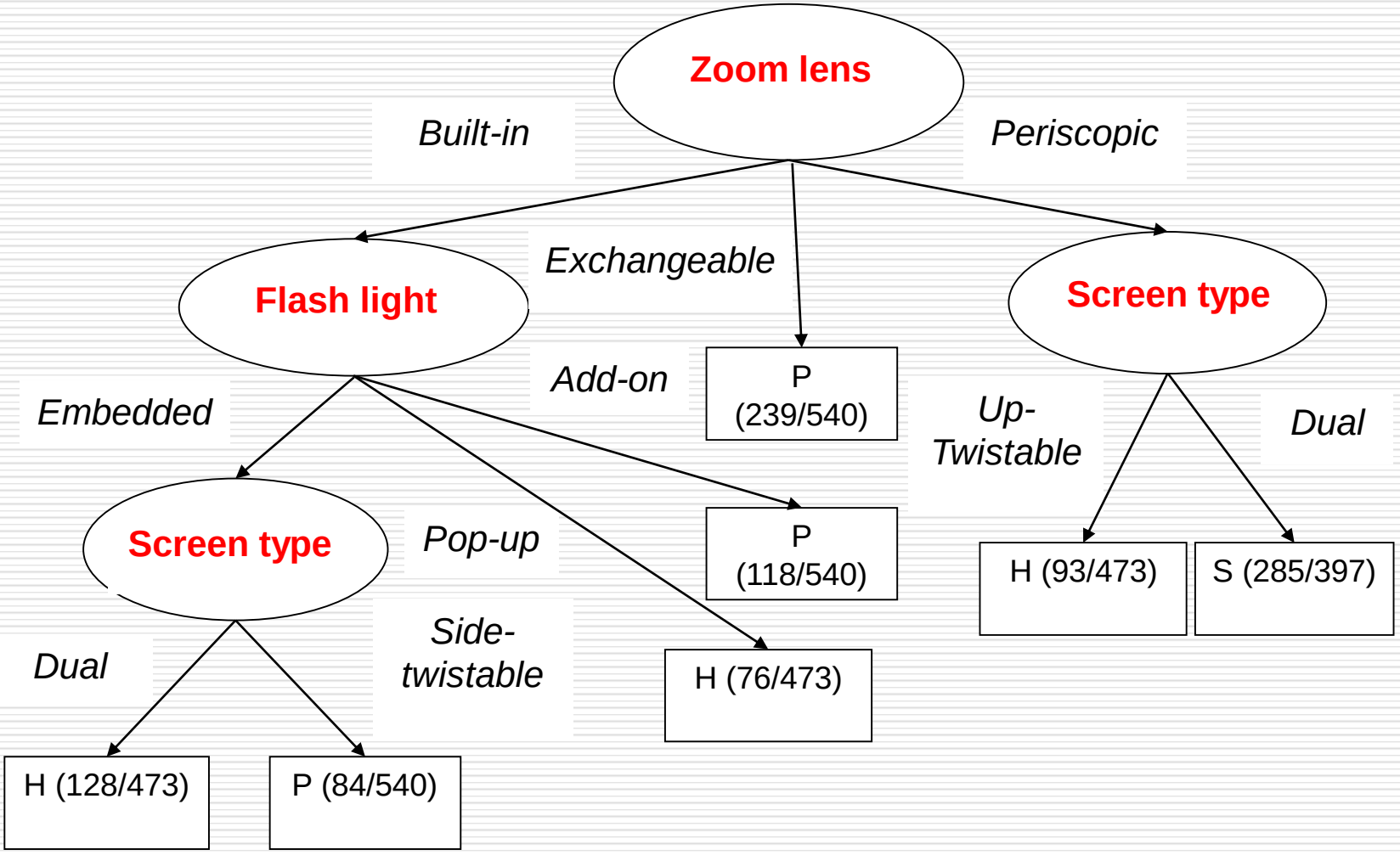


Pop-up flash-
Nikon's P600



Add-on flash

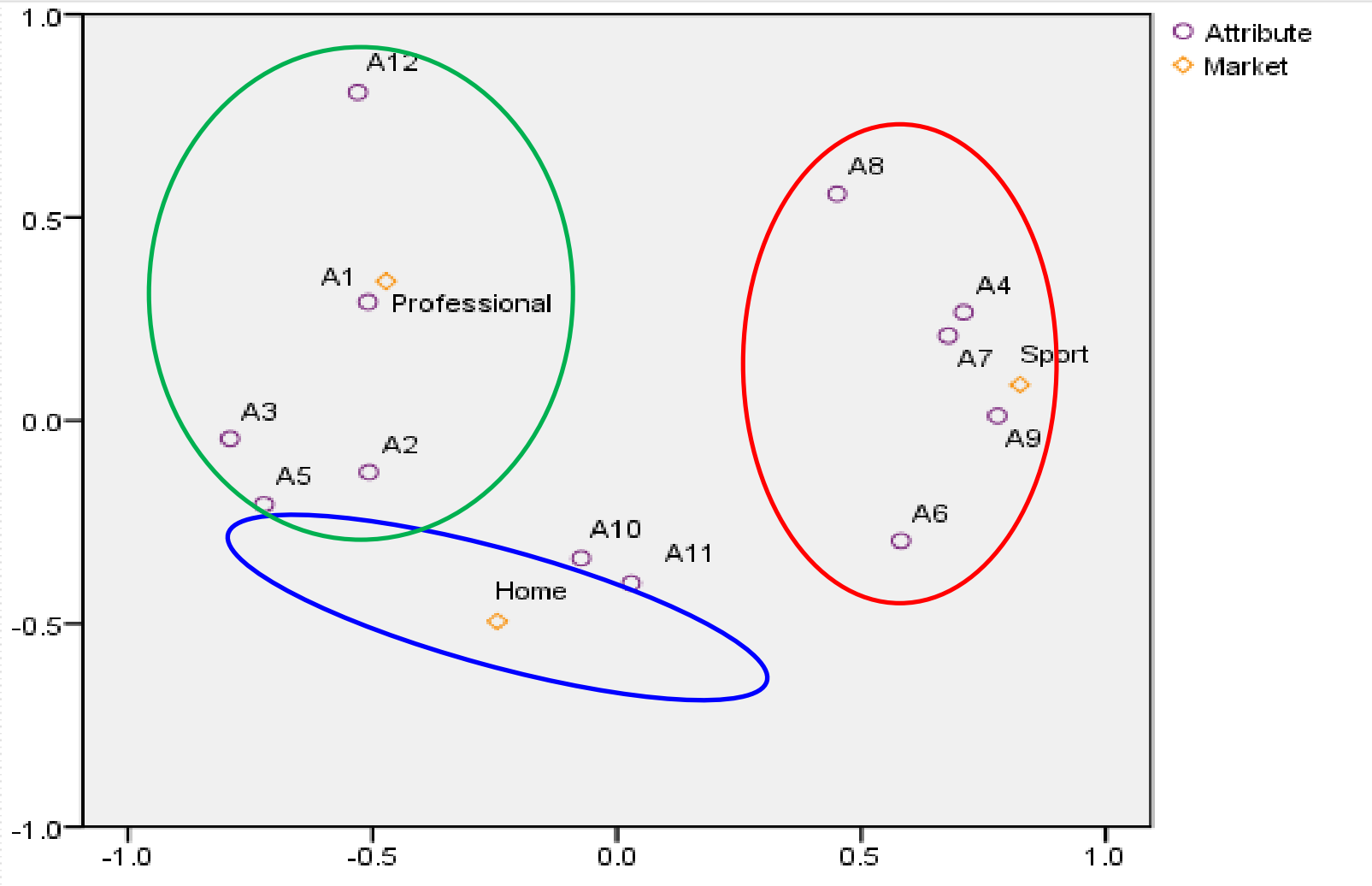
Kansei features based segmentation



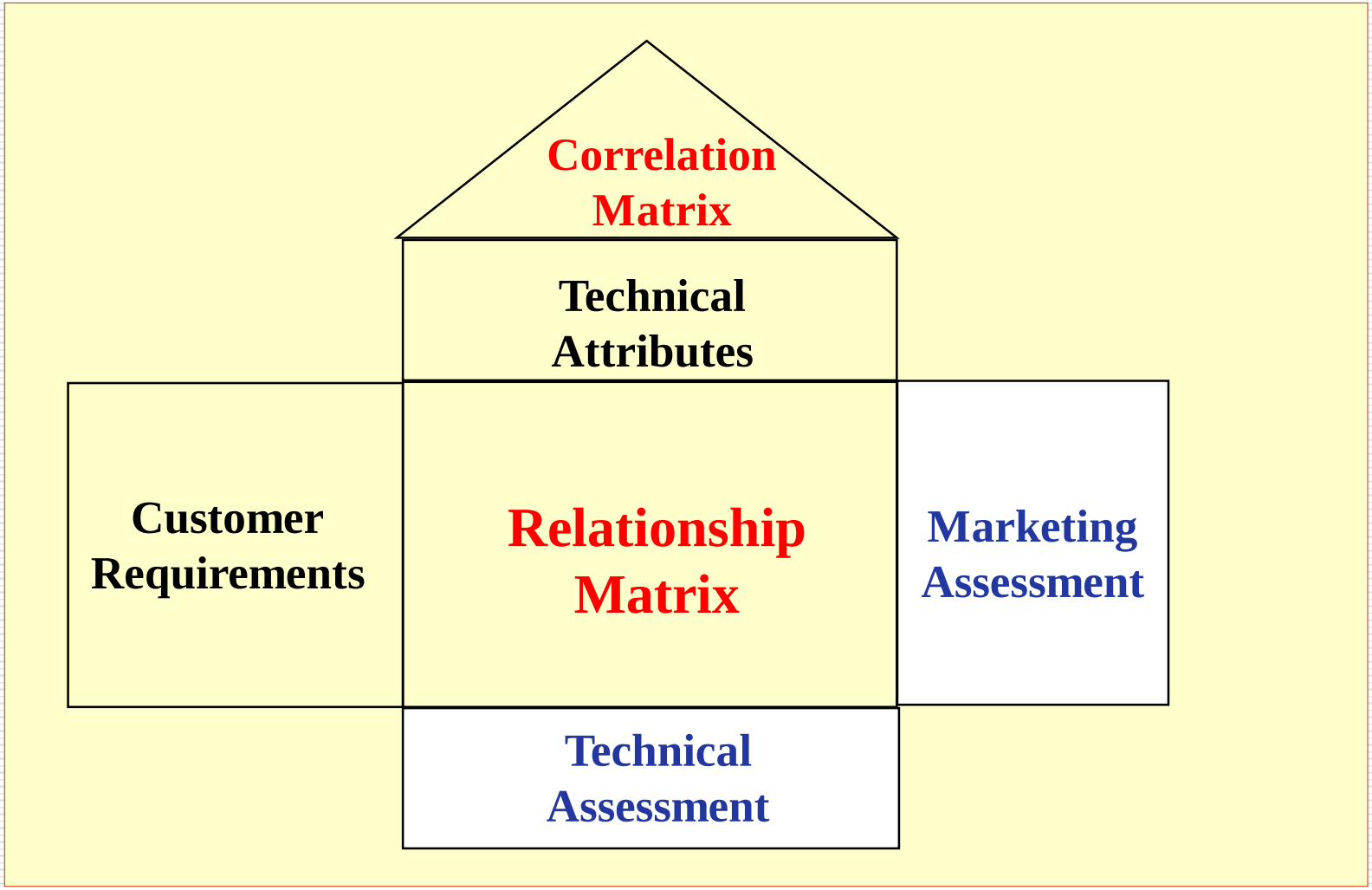
Validation assessment of decision rules

Rule	Antecedent (*if*)	Consequent (*then*)	Support= $(if \cap then)/then$	Confidence= $(if \cap then)/if$
1	(Lens=built-in) and (Flash=embedded) and (Screen=dual)	Home	27%	64.9%
2	(Lens=built-in) and (Flash=embedded) and (Screen=side-twistable)	Professional	15.6%	51.7%
3	(Lens=built-in) and (Flash=pop-up)	Home	16.1%	65.7%
4	(Lens=built-in) and (Flash=add-on)	Professional	20.7%	71%
5	(Lens=exchangeable)	Professional	44.3%	96 %
6	(Lens=periscopic) and (Screen=twistable)	Home	19.7%	54.8%
7	(Lens=periscopic) and (Screen=dual)	Sport	71.8%	82.5%

Using correspondence analysis to identify key FAs

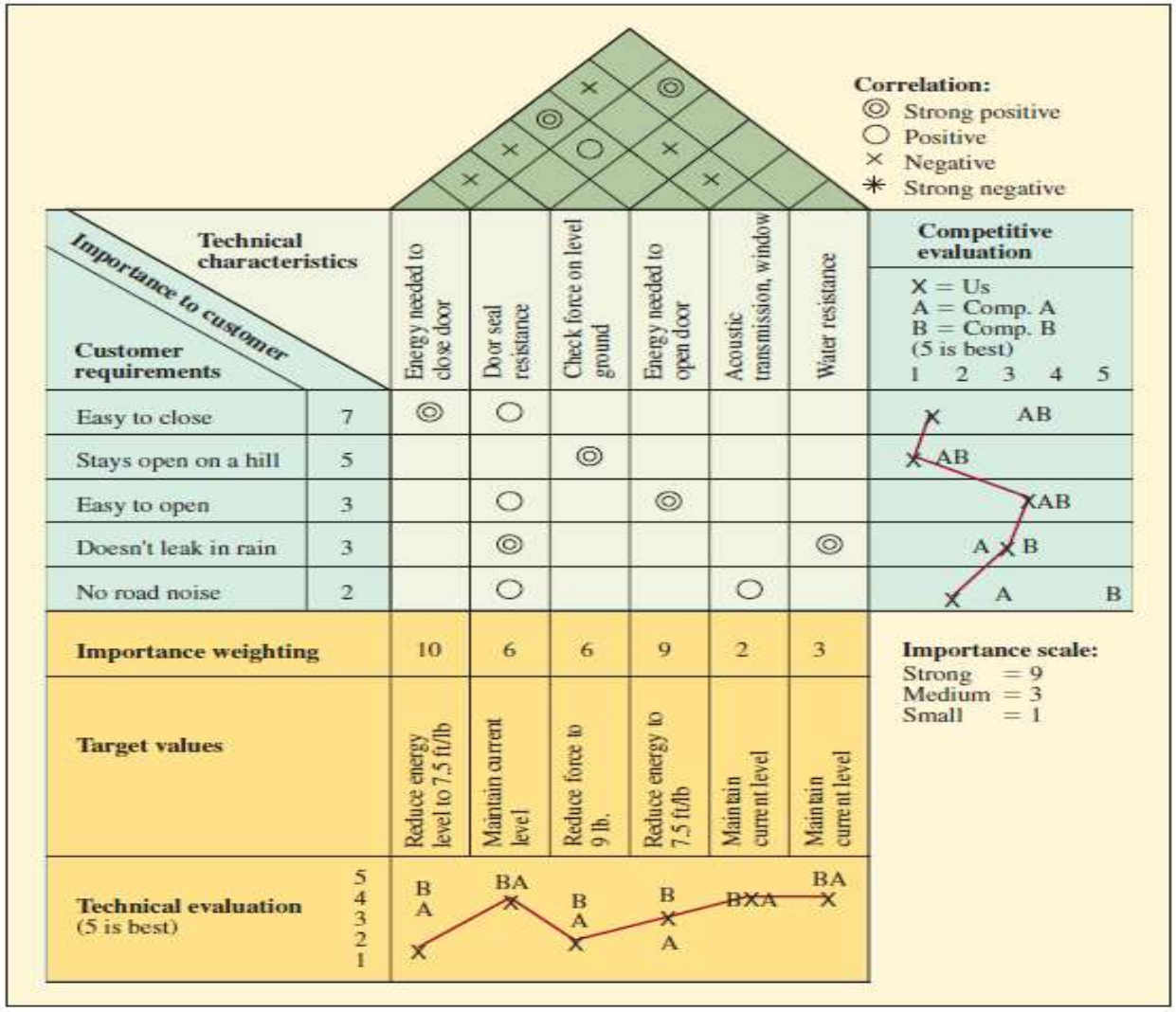


Quality Function Deployment (Akao, 1990)



House of Quality Matrix for a Car Door

Customer requirements information forms the basis for this matrix, used to translate them into operating or engineering goals



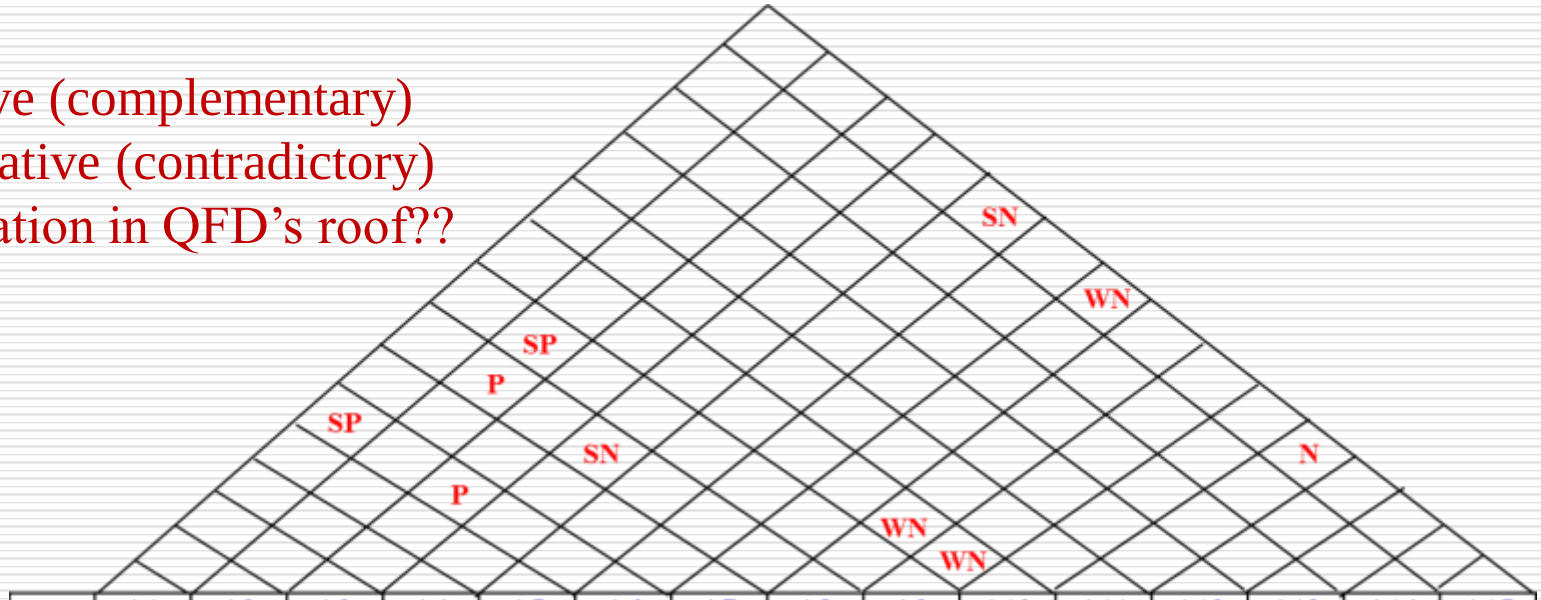
Features for smartphones

CRs	Customer requirements	TAs	Technical attributes	Ideal scale
R1	Photo quality	A1	Image sensor	↑
R2	Screen visualization	A2	Photo quality (back camera)	↑
R3	Portability	A3	Wide angle	↓
R4	Durability	A4	Screen (diagonal) size	↑
R5	System performance	A5	Screen resolution	↑
R6	Friendly user interface	A6	Selfie quality (front camera)	↑
		A7	AR/VR 3D display	**Yes/No**
		A8	Weight	↓
		A9	Thickness	↓
		A10	Water proof	**Yes/No**
		A11	CPU performance	↑
		A12	Mobile RAM capacity	↑
		A13	Touch sensitivity	↑
		A14	Gesture control	**Yes/No**
		A15	Usage hours	↑

%↑ denotes "the larger the better" while ↓ means "the smaller the better"

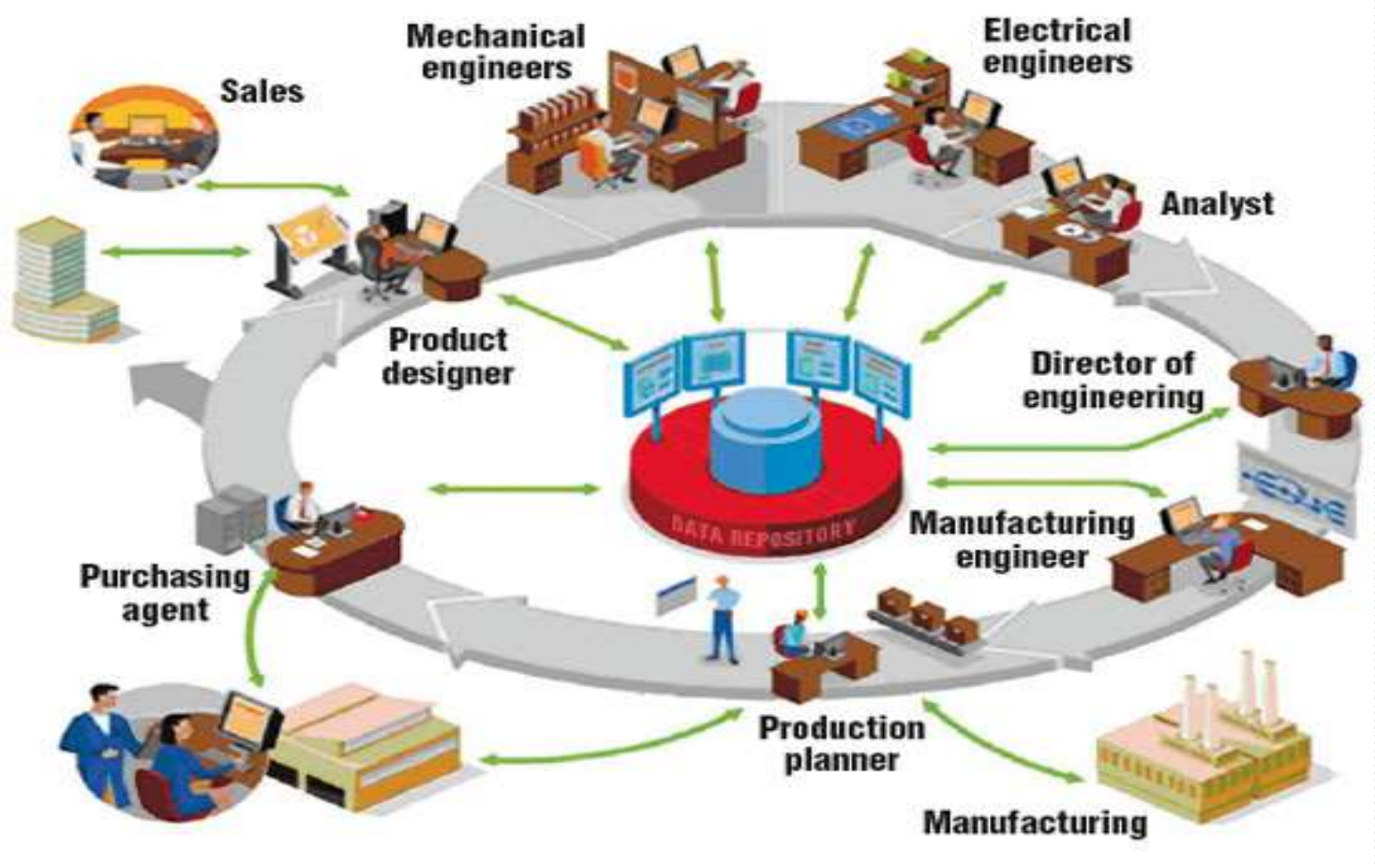
House of Quality- listening to customers

Positive (complementary) or negative (contradictory) correlation in QFD's roof??



	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12	A13	A14	A15
R1	VH	VH	VH			M									
R2				VH	H		M								
R3				VH				M	H	L					
R4				H											VH
R5											VH	H			
R6				M	H								VH	L	

Concurrent Engineering



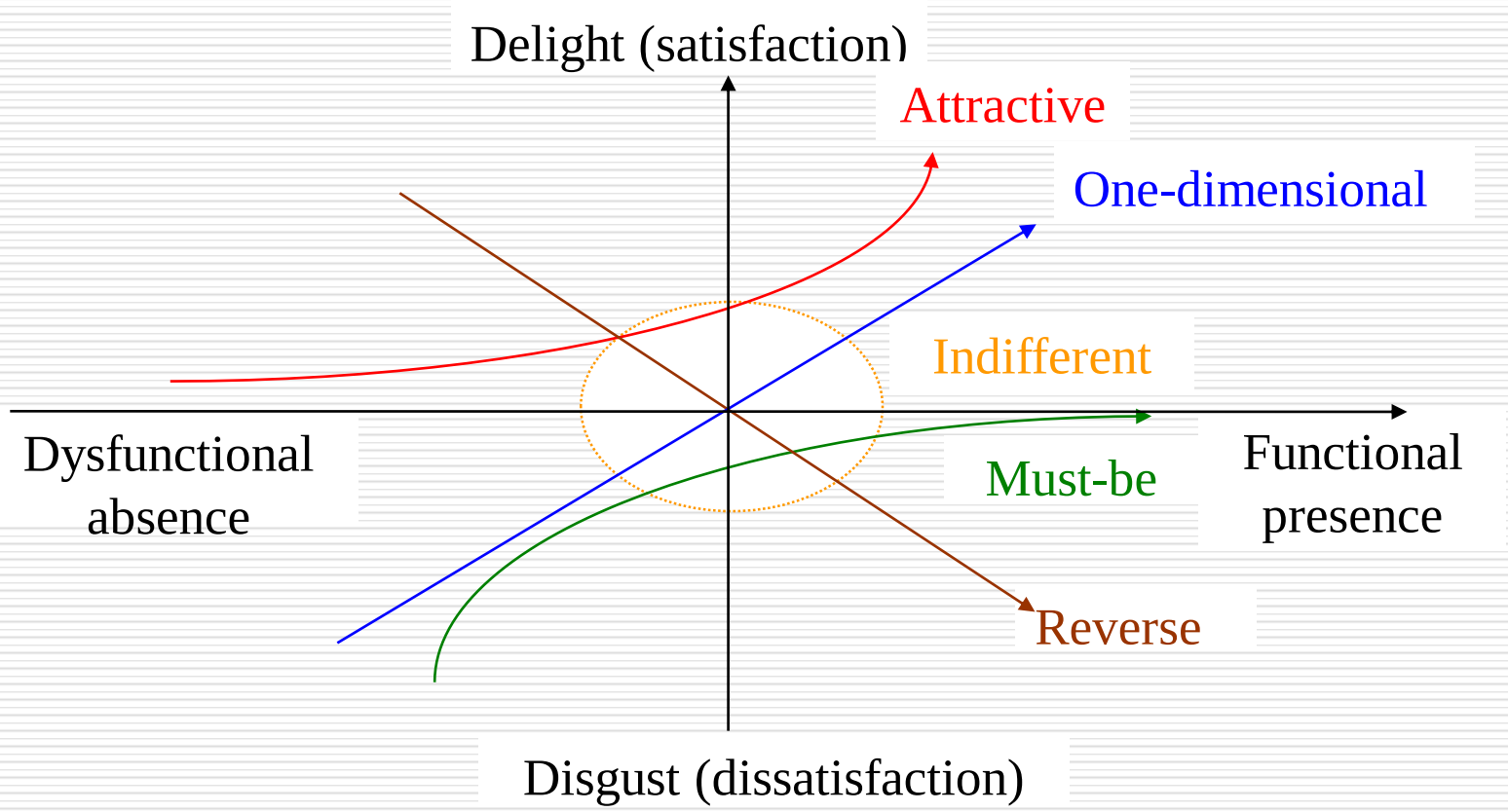
The story about Rolls-Royce? Engines? Health?



Video 2

- *A long video to show you innovative service design in 2005 by NTT DoCoMo*
- *Using QFD (quality function deployment) to transform VOC (voice of customers) into VOD (voice of designers)?*
- *How to generate innovative ideas using **TRIZ**?*
- *How to capture user perceptions using **Kano model**?*
- *Other interesting videos?*

Using Kano Model to elicit user perceptions



Analyze & Brainstorm

- Brainstorming list of features and functionality
- Determining type of product/ service requirements:
- 問卷想了解，屬性存在或不存在時的顧客感受?
- Kano Model Requirements Survey
  - User Survey
    - "Functional form" vs. "Dysfunctional Form"
      - "*How would you feel if the product **had** feature X*?"
      - "*How would you feel if the product **didn't have** feature X*?"
  - Kano Questionnaire Answers:
    - I like it (極喜愛).
    - I expect it (有期待).
    - I am neutral (沒感覺).
    - I can tolerate it (勉強忍受).
    - I dislike it (極度厭惡).

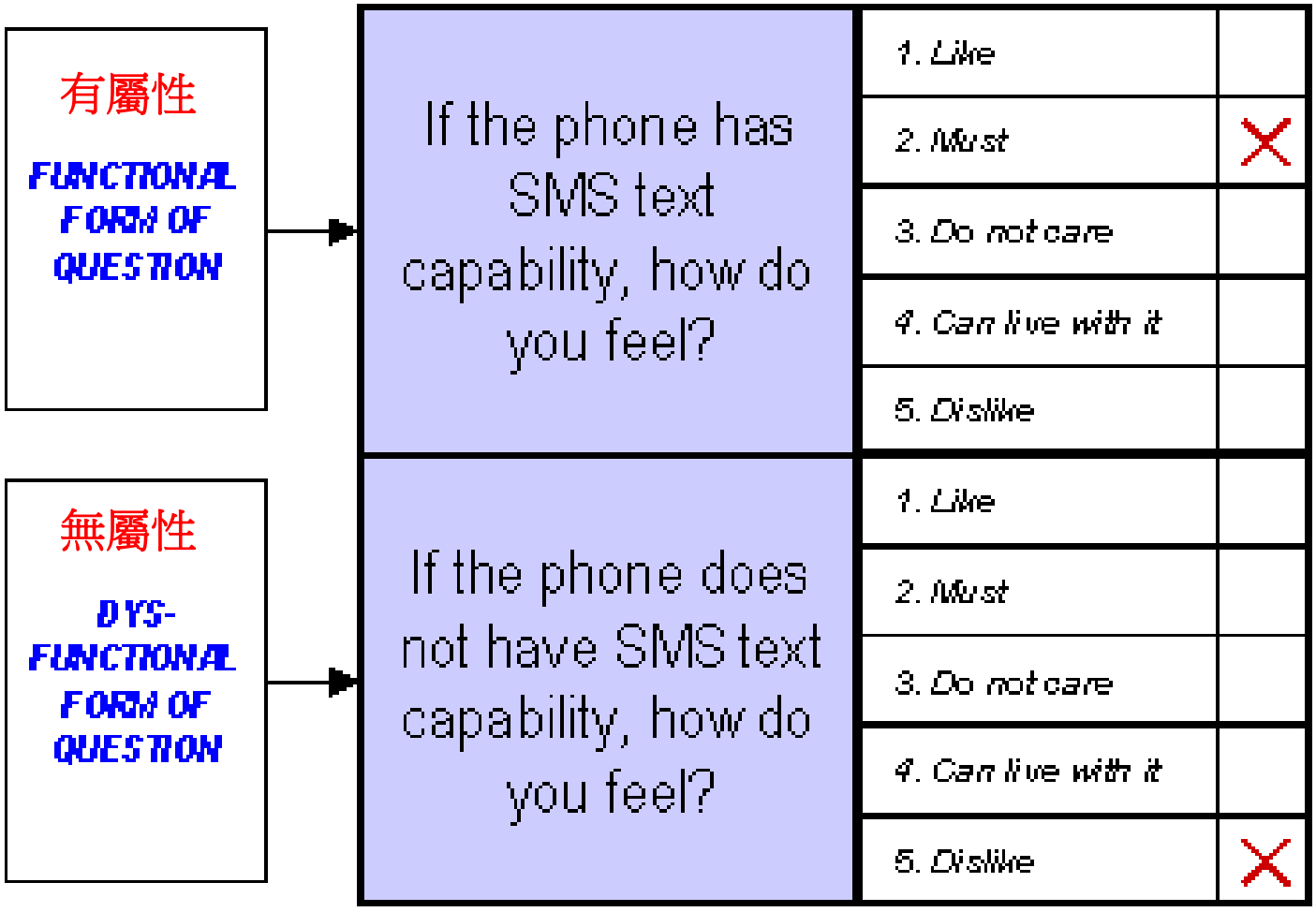
Kano問卷設計

當智慧型手機具備...		滿意	理所當然	沒感覺	能忍受	不滿意
屬性 1	正向	*				
	反向				*	
⋮	正向					
	反向					
屬性 n	正向		*			
	反向					*

問卷問項		自向題(品質要素不充足時)				
		滿意	理所當然	沒感覺	能忍受	不滿意
正向題(品質要素充足時)	滿意	Q	A	A	A	O
	理所當然	R	Q	I	I	M
	沒感覺	R	I	I	I	M
	能忍受	R	I	I	Q	M
	不滿意	R	R	R	R	Q

A代表魅力品質；M代表必需品質；R代表反轉品質；O代表一元品質；Q代表問題品質；I代表無差異品質。

An illustrated Survey



Functional vs. Dysfunctional Comparison

Customer requirements		無屬性DYS-FUNCTIONAL				
		1. Like	2. Must	3. Do not care	4. Can live with it	5. Dislike
有屬性 FUNCTIONAL	1. Like	Q	A	A	A	O
	2. Must	R	I	I	I	E M
	3. Do not care	R	I	I	I	E M
	4. Can live with it	R	I	I	I	E M
	5. Dislike	R	R	R	I	Q

O = One dimensional or linear features
I = indifferent response
A = Attractive features

E = Expected features
R = reversal (inconsistent response)
Q = Questionable response

M: must feature

Typical airline service attributes

SAs	Definitions of service attributes
A1	Appearance of crew
A2	Courtesy of crew
A3	On-board facilities (seat, space, entertainment, etc.)
A4	On-board beverage and meal
A5	Weight limit of free check-in baggage
A6	Diversity of airline shopping
A7	Weight limit of free carry-on baggage
A8	Fast check-in and boarding process
A9	Speed of handling customer requirements
A10	Skills of handling flight delays or baggage claims
A11	Extended/transferred transportation services
A12	Free services for using the VIP room at airports
A13	Records of low safety accidents
A14	Convenient departure and arrival schedules
A15	Early-bird price discount
A16	Frequent-flier mileage programs

Customer satisfaction based on Kano model

$$d_i^+ = \frac{A_i + O_i - R_i}{A_i + O_i + M_i + R_i + I_i}$$

Positive delight when attribute i is present

$$d_i^- = -\frac{O_i + M_i - R_i}{A_i + O_i + M_i + R_i + I_i}$$

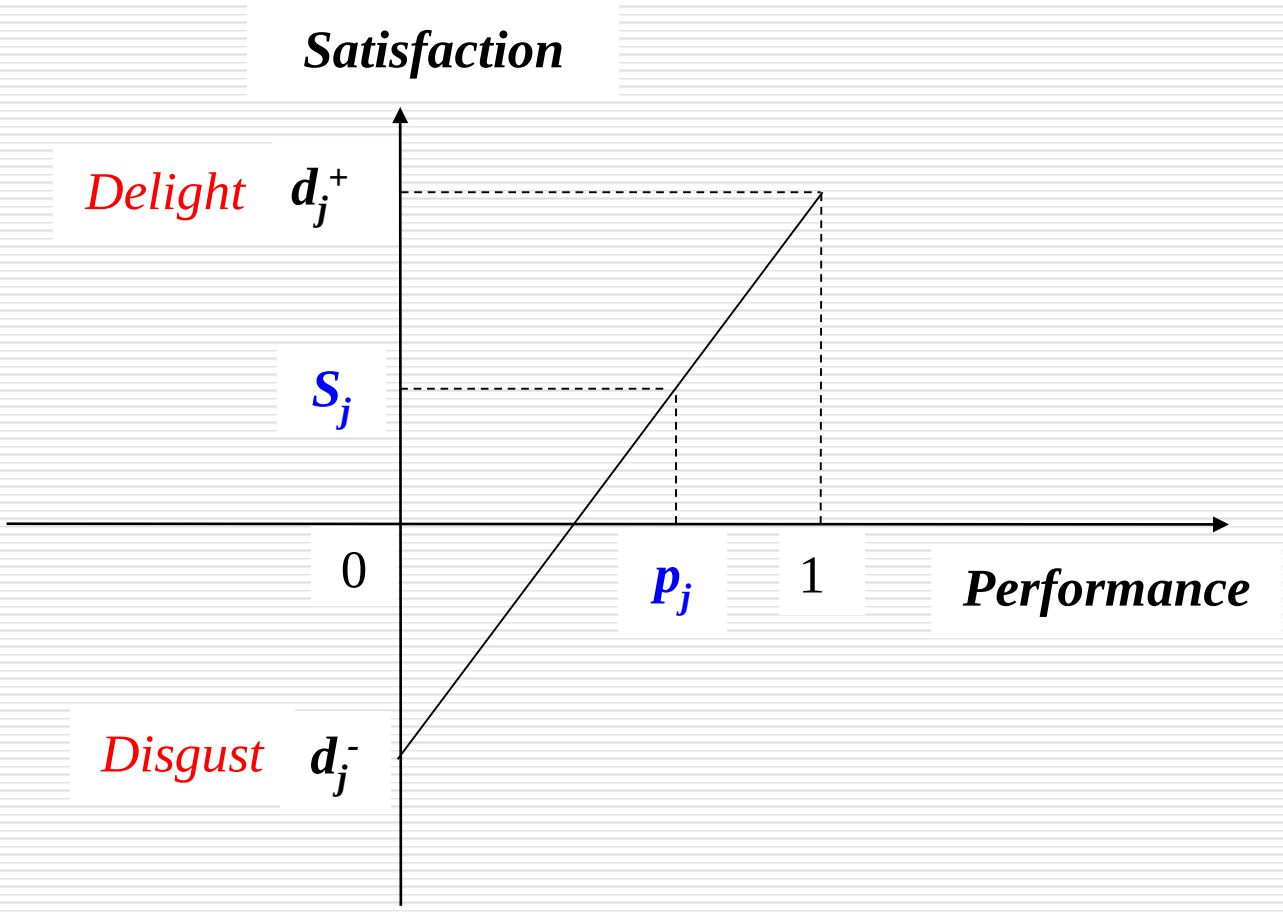
Negative disgust when attribute i is absent

$$S_j = p_j d_j^+ + (1 - p_j) d_j^-$$

Customer satisfaction with attribute j given performance

績效可由分項分數來計算 (4/5)=0.8

Converting an attribute's performance into a quantitative degree of satisfaction



Aggregated Kano categories (in percentages)

SAs	A	M	O	R	I	Q	Delight	Disgust
A1	40%	10%	7%		43%		0.47	-0.17
A2	9%	44%	45%		2%		0.54	-0.89
A3	45%	6%	25%		24%		0.7	-0.31
A4	52%	10%	25%		13%		0.77	-0.35
A5	15%	28%	54%		3%		0.69	-0.82
A6	32%		4%		64%		0.36	-0.04
A7	26%	3%	4%		67%		0.3	-0.07
A8	9%	31%	54%		6%		0.63	-0.85
A9	11%	22%	62%		5%		0.73	-0.84
A10	4%	49%	40%		7%		0.44	-0.89
A11	16%	27%	38%		19%		0.54	-0.65
A12	61%		4%		35%		0.65	-0.04
A13	10%	34%	45%		11%		0.55	-0.79
A14	25%	33%	29%		13%		0.54	-0.62
A15	53%	3%	14%		30%		0.67	-0.17
A16	69%	2%	8%		21%		0.77	-0.1

Smart products

Hapifork 智慧叉 (666) Hapifork, Eat Slowly - Lose Weight - Feel Great - YouTube

Zepp 智慧球拍 (666) Zepp Labs | Golf | Swing Analyzer - YouTube

Vessyl 智慧杯 (666) Smart Cup "Vessyl" - Automatically Track Everything You Drink - YouTube

Beam toothbrush 智慧牙刷 (666) Bluetooth Enabled Toothbrush Tracks Your Dental Habits - YouTube

Sensoria socks 智慧襪 (666) Sensoria Smart Socks, coaching runners - YouTube

產品組合管理

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